

Tianjian Qin, PhD

Computational Biologist & Data Scientist

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🔗 [HerdLink.nl](https://herdlink.nl) | 🌐 [EvoLandEco](https://evolandeco.nl) | 📄 [tjqin](https://tjqin.nl)

// Research Profile

- I develop machine learning and deep learning methods, statistical models, simulation frameworks, and research software for complex biological systems. My work spans evolutionary biology, ecology, infectious disease epidemiology, and One Health research. I work with domain experts to turn scientific questions into transparent models, scalable pipelines, and usable tools.

// Research Experience

Postdoctoral Researcher

2025–present

Wageningen University & Research, Infectious Disease Epidemiology Group

- Develop generative and predictive models for dynamic livestock movement networks, including neural models for spatial point patterns, Transformer models for link forecasting, stochastic block models, and transmission simulations.
- Build reproducible HPC pipelines that integrate movement records, farm metadata, spatial information, temporal covariates, and epidemiological context.
- Translate empirical data and stakeholder questions into modelling workflows, open software, visual analysis, and decision support outputs across animal and public health domains.

Doctoral Researcher in Theoretical & Computational Biology

2019–2026

University of Groningen, Groningen Institute for Evolutionary Life Sciences

- Developed the EVE stochastic birth-death model and high-performance simulation and inference methods for diversification dependent on evolutionary relatedness.
- Designed EvoNN, an ensemble learning framework combining graph, recurrent, and dense neural networks for parameter inference from phylogenetic trees.
- Built and maintained reproducible research software in R, C++, and Python, with extensive CPU and GPU computing, testing, documentation, and open releases.

MSc Researcher in Wetland Ecology & Invasion Biology

2016–2019

Beijing Forestry University, School of Nature Conservation

- Conducted field surveys across 18 Chinese provinces and received training in controlled experiments, GIS, phylogenetic metrics, community and trait analyses, spatial analysis, and environmental data analysis.
- Built automated image analysis and phylogenetic reconstruction workflows and published quantitative studies of invasion resistance and community assembly.

// Research Funding & Proposal Development

NextdAI: Integrated AI-Driven Early Detection System

IN DEVELOPMENT

NWO KIC Early Detection of (Emerging) Infectious Disease Outbreaks

2026

- Co-developing a novel AI architecture and a data infrastructure plan for heterogeneous surveillance streams, anomaly detection, neuro-symbolic characterisation, forecasting, and knowledge-based decision support.

SSS-mod: Preparedness Package for Future Respiratory Outbreaks

AWARDED

ZonMw Implementatie-impuls COVID-19 programma

Mar–Dec 2026

- Project team member. Responsible for programming for data preparation, recalibration, model output generation, and interface development, supporting the reuse of a setting-specific transmission model during future respiratory outbreaks.

- › Compute allocation supporting data-intensive simulation, neural inference, and model evaluation on high-performance computing systems.

// Selected Research Projects

HerdLink.nl

LIVE PLATFORM

<https://herdlink.nl>

Developer | JavaScript | D3.js

Developed and deployed a public web platform for exploring anonymised and aggregated Dutch livestock trade networks. HerdLink links graph and map views with node focus, temporal replay, network summaries, and transmission simulation for research exploration, communication, and education.

EUPAHW HEV: transmission pathways across pigs, wild boar, and humans

Researcher

European Partnership on Animal Health and Welfare

- › Developing research questions and analysis strategies that join phylodynamics, animal movement networks, wildlife and landscape information, and intervention timing. The work targets transmission direction, spillover between species, long-range lineage movement, and generalisable One Health surveillance design.

NetForge: dynamic livestock network modelling

Developer

Python | network science | epidemic simulation

- › Developing a research toolkit for movement network synthesis, statistical evaluation, metadata generation, and transmission experiments. Current work includes physics-informed network generators and models for spatial and temporal link prediction.

EVE & EvoNN: simulation and neural inference from phylogenies

Researcher

R | C++ | Python | GPU computing

- › Created stochastic models, simulators, and ensemble neural networks that learn jointly from graph structure, branching times, and summary statistics. The work supports peer-reviewed publications, reusable packages, and a completed doctoral thesis.

// Teaching & Mentoring

- › Teaching assistant for annual coding assignments and computer practicals in island biogeography at the University of Groningen.
- › Guest lectures, tutorials, and seminars on stochastic processes, diversification models, deep learning, neural networks, and network statistics.
- › Lectures and workshops at Wageningen University & Research and Wageningen Bioveterinary Research on phylodynamics, birth-death models, and network modelling.
- › Pair programming sessions on test-driven development and collaborative software design.
- › Informal mentoring of BSc and MSc students in biostatistics, data visualisation, and scientific writing, and of junior PhD researchers in planning and day-to-day research practice.

// Education

PhD in Theoretical Biology

2019–2026

University of Groningen, Netherlands

- › Thesis: *Diversification Models and Neural Inference*. |  [Thesis PDF](#)
- › Supervisors: Prof. Dr. Rampal Etienne, Prof. Dr. Luis Valente, and Dr. Koen van Benthem.

MSc in Wetland Ecology

2016–2019

Beijing Forestry University, China

- › Supervisors: Prof. Dr. Hongli Li and Prof. Dr. Feihai Yu.

BSc in Marine Biology

2012–2016

Nanjing Normal University, China

// Publications

PEER-REVIEWED ARTICLES

1. **Qin, T.**, van Benthem, K., Valente, L.[†], & Etienne, R. S.[†] (2026). Parameter estimation from phylogenetic trees using neural networks and ensemble learning. *Systematic Biology*, 75(2), 344–365.
2. **Qin, T.**, Valente, L.[†], & Etienne, R. S.[†] (2025). Impact of evolutionary relatedness on species diversification and tree shape. *Journal of Theoretical Biology*, 598, 111992.
3. Sun, K., Liu, X.-S., **Qin, T.-J.**, Jiang, F., Cai, J.-F., Shen, Y.-L., A, S.-H., & Li, H.-L. (2021). Relative abundance of invasive plants more effectively explains the response of wetland communities to different invasion degrees than phylogenetic evenness. *Journal of Plant Ecology*.
4. **Qin, T.**, Zhou, J., Sun, Y., Müller-Schärer, H., Luo, F., Dong, B., Li, H., & Yu, F.-H. (2020). Phylogenetic diversity is a better predictor of wetland community resistance to *Alternanthera philoxeroides* invasion than species richness. *Plant Biology*.
5. Wan, J.-Z., Wang, M.-Z., **Qin, T.**, Bu, X.-Q., Li, H.-L., & Yu, F.-H. (2019). Spatial environmental heterogeneity may be the driver of functional trait variation in *Hydrocotyle vulgaris*, an aquatic plant invader. *Aquatic Biology*.
6. **Qin, T.-J.***, Guan, Y.-T.*, Quan, H., Dong, B.-C., Luo, F.-L., Zhang, M.-X., Li, H.-L., & Yu, F.-H. (2019). Growth traits of the exotic plant *Hydrocotyle vulgaris* and the evenness of resident plant communities are mediated by community age, not species diversity. *Weed Research*.
7. **Qin, T.-J.**, Guan, Y.-T., Zhang, M.-X., Li, H.-L., & Yu, F.-H. (2018). Sediment type and nitrogen deposition affect the relationship between *Alternanthera philoxeroides* and experimental wetland plant communities. *Marine and Freshwater Research*.
8. Liu, L., Guan, Y.-T., **Qin, T.-J.**, Wang, Y.-Y., Li, H.-L., & Zhi, Y.-B. (2018). Effects of water regime on the growth of the submerged macrophyte *Ceratophyllum demersum* at different densities. *Journal of Freshwater Ecology*.

PREPRINT

1. **Qin, T.**, van Benthem, K., Valente, L.[†], & Etienne, R. S.[†] (2026). Identifying evolutionary relatedness effects on diversification from phylogenies using neural networks. *bioRxiv*.

CONFERENCE

1. **Qin, T.**, Atamer Balkan, B., Schmid, B., & ten Bosch, Q. Generating livestock movement networks that preserve epidemic behaviour with physics informed machine learning. *ModAH Conference 2026, Nantes, France*.

* Equal contribution. [†] Joint senior authors.

// Honours & Awards

- › **Denise Kirschner Best Student Paper Prize** , *Journal of Theoretical Biology* (2025). Inaugural prize for graduate student research.
- › **First-Class Academic Performance Award**, Beijing Forestry University (2017–2018 and 2018–2019).
- › **Graduate Academic Innovation Award**, Beijing Forestry University (2018).

// Academic & Community Service

- › **Peer reviewer** for *Systematic Biology*, *PLOS Computational Biology*, *Epidemics*, *BMC Public Health*, and *BMC Plant Biology*.
- › **Contributor** to various GitHub open-source repositories.
- › **Vice President**, Graduate Student Council, School of Nature Conservation, Beijing Forestry University (2017–2018).
- › **Student representative**, Wetland Ecology Group, Beijing Forestry University (2016–2019).

// Research Software

NetForge

Developer

Movement network modelling and transmission simulation | Python

- › Command-line tools for network synthesis, metadata generation, evaluation, and epidemic experiments.

EvoNN & evesim

Developer

Neural inference and stochastic simulation | R/Python/C++

- › Pretrained neural inference from phylogenies and fast simulation of evolutionary relatedness dependent birth-death processes.

NetSpectra, EvoLab, & miniape

Developer

Interactive scientific tools | JavaScript

- › Network dynamics, lineage simulation, tree reconstruction, and browser-based phylogenetic data handling.

treestats, DDD, & DAISIE

Collaborator

High-performance research packages | R/C++

- › Contributions to data structures, phylogenetic summary statistics, and simulation and inference for diversification and island biogeography models.

// Technical Expertise

Machine learning	Neural networks, transformer models, ensemble learning, and benchmarking.
Statistics & inference	Stochastic processes, simulation-based inference, and likelihood methods.
Programming	Python, R, C/C++, JavaScript, SQL; package design, analysis pipelines, and interactive visualisation.
Research computing	Linux, Bash, HPC, data infrastructure, Docker, Git, GitHub workflows, agentic coding, and CI/CD.
Communication	Scientific writing, $\text{\LaTeX}$, TikZ, PGFPlots, D3.js, and interface design for domain experts and students.

// Languages

Chinese Native | English Proficient | Dutch Basic | German Basic